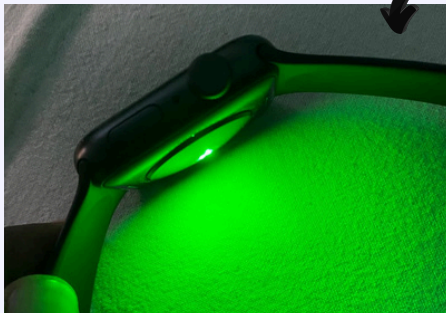


WEARABLE BIOSENSORS AND EARLY DIAGNOSIS DETECTION: TRANSFORMING PREVENTIVE HEALTHCARE

THE TOOLS BEHIND IT

Photoplethysmography (PPG) is a noninvasive optical tool that measures blood volume changes in vessels to provide information about heart rate and other measures related to vascular health. It has been incorporated into many commercial wearable devices because of its simplicity, comfort, and cost-effectiveness. Devices incorporating photoplethysmography can be used to diagnose arrhythmias and monitor oxygen saturation levels.



It uses those little green lights on the skin-side of the wearable to track the amount of blood passing through your wrist at any given time. Clip-on pulse oximeters used by doctors are the same type of tech. The addition of PPG to a wrist tracker allows for the measurement of raw data like heart rate and breathing rate. From this data, the wearable can estimate a number of physiological metrics, including sleeping stages.

PARKINSON'S DISEASE

FDA-approved wearable devices are improving Parkinson's disease care by providing continuous, objective monitoring of motor symptoms in daily life. Devices such as PDMonitor, Personal KinetiGraph, STAT-ON™, and KinesiaONE™/Kinesia360™ use body-worn sensors and data analytics to track symptoms like gait changes, tremor, bradykinesia, dyskinesia, freezing of gait, and medication effects, enabling early detection of fluctuations, safer monitoring, and more personalised, data-driven treatment decisions.



Sources: <https://www.sciencedirect.com/science/article/abs/pii/S0263224125012977>

<https://www.mdpi.com/2079-9292/12/13/2923>

<https://theconversation.com/how-do-sleep-trackers-work-and-are-they-worth-it-a-sleep-scientist-breaks-it-down-258304>

<https://www.mdpi.com/2224-2708/14/2/23>